

Week 13 Homework: Decisions, Games, and Uncertainty

Goal: connect Pareto optimization and Nash equilibrium to probability, p-values, and scientific experiments.

AI policy. You may use AI or web search to understand official documentation, ask conceptual questions, or interpret an error message. You may **not** ask AI to write the required Pareto logic, Nash-equilibrium logic, probability calculations, p-value simulation, experiment analysis, or final explanation. The TODO code and reasoning must be your own manual work. If you use AI, add one sentence stating what it helped you understand.

Key reminders. For two minimization objectives, plan u dominates plan v if u is no worse in both objectives and strictly better in at least one. A pure-strategy Nash equilibrium is a cell where both players are best responding. For a discrete random variable,

$$E[X] = \sum_x xP(X = x), \quad \text{Var}(X) = \sum_x (x - E[X])^2 P(X = x).$$

A p-value is the probability, assuming H_0 , of a test statistic at least as extreme as the observed one.

Monday: Pareto Frontier and Weighted Decisions

Lower cost and lower emissions are better.

Plan	Cost	Emissions
A: fossil-heavy	35	95
B: balanced	50	60
C: renewable-heavy	80	25
D: inefficient backup	90	90

1. List every pair of plans for which one dominates the other. Explain each comparison.
2. Identify the Pareto-optimal plans and sketch cost versus emissions. Mark the frontier.
3. Calculate $J = \lambda(\text{cost}) + (1 - \lambda)(\text{emissions})$ for every plan when $\lambda = 0.2, 0.6$, and 0.8 . State the winner for each weight.
4. Explain why the Pareto frontier does not automatically identify one best plan.
5. Suppose cost is later written in dollars as 35,000, 50,000, 80,000, and 90,000 while emissions keep their original values. Explain why using the same weighted sum would be misleading and how normalization helps.

Create a Python file named `week13_probability_games.py`. Begin with the following code and complete every TODO manually.

```
import numpy as np

plans = [
    {"name": "A", "cost": 35.0, "emissions": 95.0},
    {"name": "B", "cost": 50.0, "emissions": 60.0},
    {"name": "C", "cost": 80.0, "emissions": 25.0},
    {"name": "D", "cost": 90.0, "emissions": 90.0},
]

def is_dominated(plan, all_plans):
    for other in all_plans:
        if other is plan:
            continue
        # TODO: Is other no worse in both objectives?
        no_worse = False
        # TODO: Is other strictly better in at least one objective?
        strictly_better = False
        # TODO: Return True when both conditions hold.
        # TODO: No dominating plan was found.
        pass

pareto_names = []
for plan in plans:
    # TODO: Append the name only when plan is not dominated.
    pass

print("Pareto-optimal plans:", pareto_names)
```

Tuesday: Prisoner's Dilemma and Nash Equilibrium

Payoffs are negative years in prison, so larger values are better. Rows belong to A; columns belong to B. Strategy 0 means stay silent and strategy 1 means confess.

$$A = \begin{bmatrix} -1 & -5 \\ 0 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 0 \\ -5 & -3 \end{bmatrix}.$$

1. Find A's best response to each of B's strategies.
2. Find B's best response to each of A's strategies.
3. Identify each player's dominant strategy and the Nash equilibrium.
4. Verify the equilibrium by checking both possible unilateral deviations.
5. Identify the socially better outcome. Explain why it is Pareto superior but not stable.

Add the following section to your Python file.

```
A = np.array([[ -1.0, -5.0],
              [ 0.0, -3.0]])
B = np.array([[ -1.0,  0.0],
              [-5.0, -3.0]])

def find_pure_nash(A, B):
    equilibria = []
    for row in range(A.shape[0]):
        for col in range(A.shape[1]):
            # A may switch rows while B's column stays fixed.
            # TODO: Test whether A[row, col] is maximal in column col.
            a_best = False
            # B may switch columns while A's row stays fixed.
            # TODO: Test whether B[row, col] is maximal in row row.
            b_best = False
            # TODO: Append (row, col) if both are best responding.
    return equilibria

print("Pure Nash equilibria:", find_pure_nash(A, B))
```

Now analyze this demand-response game:

	B shifts	B uses peak
A shifts	(3, 3)	(1, 4)
A uses peak	(4, 1)	(2, 2)

Find the Nash equilibrium and Pareto improvement. Then add 2 payoff points to each player whenever that player shifts and write the new matrix. Find its equilibrium and explain how the incentive changes behavior.

Wednesday: Probability and Comparing Random Variables

Plan X has cost 20 with probability 0.60 and cost 80 with probability 0.40. Plan Y always costs 45.

1. Compute $E[X]$, $E[Y]$, $\text{Var}(X)$, $\text{Var}(Y)$, and both standard deviations.
2. Compute $P(X < Y)$, $P(X > Y)$, and $P(X > 50)$.
3. Which plan has lower expected cost? Which has lower risk? Give one situation in which a grid operator could reasonably choose each plan.
4. Explain why the statement “ X is smaller than Y ” is incomplete when X and Y are random variables.

Use this table of 100 observed days.

	Overload	No overload	Total
Hot	8	12	20
Not hot	2	78	80
Total	10	90	100

Calculate $P(\text{hot})$, $P(\text{overload})$, $P(\text{hot and overload})$, $P(\text{overload} \mid \text{hot})$, and $P(\text{overload} \mid \text{not hot})$. Are hot weather and overload independent? Support the answer numerically.

Add this simulation to your Python file.

```
rng = np.random.default_rng(13)
n = 100_000
X = rng.choice([20.0, 80.0], size=n, p=[0.6, 0.4])
Y = np.full(n, 45.0)

# TODO: Print the sample mean of X and Y.
# TODO: Print the sample standard deviation of X and Y.
# TODO: Estimate  $P(X < Y)$  with a Boolean comparison and np.mean.
# TODO: Estimate  $P(X > 50)$ .
```

Compare every simulated result with Wednesday's hand calculation. Explain why the values are close but not exactly equal.

Thursday: Hypothesis Test and Simulated p-Value

A researcher suspects that a coin favors heads. The hypotheses are

$$H_0 : p = 0.5, \quad H_1 : p > 0.5.$$

The researcher observes 9 heads in 10 flips.

1. Explain why this is a one-sided test.
2. Under H_0 , calculate the exact p-value $P(X \geq 9)$ using

$$P(X = k) = \binom{10}{k} (0.5)^{10}.$$

3. Use $\alpha = 0.05$. State the decision and conclusion in context.
4. Write one sentence explaining why the p-value is not the probability that H_0 is true.
5. If only 8 heads had appeared, calculate $P(X \geq 8)$ and state the new decision.

Complete the simulation.

```
rng = np.random.default_rng(13)
n_simulations = 100_000

# Simulate the number of heads in 10 fair-coin flips.
# TODO: Use rng.binomial and store the array as null_counts.

observed_heads = 9
# TODO: Estimate  $P(\text{null\_counts} \geq \text{observed\_heads})$ .
p_value = None

print("simulated one-sided p-value:", p_value)
print("exact one-sided p-value:", 11 / 1024)

# TODO: Repeat with 1,000 simulations and 1,000,000 simulations.
# Explain how the number of simulations affects stability.
```

Friday: Design a Scientific Experiment

A grid operator wants to know whether a new text message reduces household electricity use from 5–8 p.m. Design an experiment that could support a causal conclusion.

1. State the population, sample, treatment, control, and primary outcome.
2. Explain how households will be randomly assigned.
3. State H_0 and H_1 in words and symbols. Say whether the test is one-sided or two-sided and justify the choice.
4. Name two possible confounders or sources of bias and explain how the design addresses them.

5. State what effect size would be practically important before collecting data.
6. Write a conclusion template for each possible result: $p\text{-value} \leq 0.05$ and $p\text{-value} > 0.05$. Use “reject” or “fail to reject,” not “prove” or “accept.”
7. Explain why a small p-value alone would not prove that the program is useful.

Feynman Technique post. Write a beginner-friendly learning note titled “From Best Decisions to Evidence.” It must:

1. explain a Pareto frontier with the cost–emissions example;
2. explain Nash equilibrium with the prisoner’s dilemma;
3. explain why random variables cannot always be ranked by one observed number;
4. define a p-value correctly without saying it is the probability that H_0 is true;
5. include one hand calculation, payoff matrix, plot, or screenshot from your work; and
6. end with one question you still have about uncertainty or scientific experiments.

Submission checklist. Monday Pareto analysis and code; Tuesday game analysis, incentive redesign, and code; Wednesday probability calculations and simulation; Thursday exact and simulated p-values; Friday experiment plan; Feynman Technique post; one sentence disclosing any permitted AI help.